

DIGITAL FORENSICS CASE STUDY

Course: Information Security / Digital Forensics

Level: Undergraduate

Case Title: Unauthorized Access and Data Theft Investigation

1. Case Background

ABC Pvt. Ltd. is an IT company that recently detected suspicious activity on an employee's computer system. The employee, Ali, reported that he did not access his system during late-night hours, yet system logs indicate otherwise. The organization suspects unauthorized access and possible data theft involving sensitive company files.

2. Incident Description

Date of Incident: March 10

Time: 11:30 PM – 2:00 AM

System: Windows-based computer

User: Ali

3. Observations

- Multiple failed and successful login attempts
- Access to confidential files outside working hours
- USB device connected during the suspicious time
- Antivirus software was disabled
- Files copied and later deleted
- Internet browsing activity involving file-sharing websites

4. Available Evidence

- System event logs
- File metadata (Created, Modified, Accessed timestamps)
- USB connection history
- Browser history
- Recovered deleted files

5. Student Task

You are a digital forensic investigator assigned to analyze this case. Examine the scenario and answer the following questions based on forensic principles.

6. Questions

Section A: Basic Concepts

1. Identify the type of cybercrime involved in this case.
2. Explain the role of digital forensics in investigating such incidents.
3. List possible sources of digital evidence in this scenario.

Section B: Log Analysis

4. What information can be obtained from system event logs?
5. How can login activity help identify suspicious behavior?
6. What is the significance of late-night system access?

Section C: File System Analysis

7. Define file metadata.
8. How do timestamps (Created, Modified, Accessed) assist in investigation?
9. What can be concluded if files are accessed outside working hours?

Section D: USB and Data Transfer

10. Why is USB history important in forensic investigations?
11. What evidence suggests data transfer to an external device?
12. How can data exfiltration be confirmed?

Section E: Anti-Forensics

13. What does disabling antivirus software indicate?
14. Define anti-forensics and give two examples.
15. Why might an attacker delete files after copying them?

Section F: Analysis and Conclusion

16. Based on the case, is this likely an insider or outsider attack? Justify your answer.
17. What additional evidence would strengthen the investigation?
18. What actions should the company take after this incident?
19. Suggest at least three preventive security measures.

7. Optional Task (Bonus)

Create a timeline of events based on the case

Recommend tools and techniques for deeper forensic investigation